

Question ID: 13e5a5d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question

$5|x| = 45$

What is the positive solution to the given equation?

Correct Answer: 9

Rationale

The correct answer is 9. Dividing both sides of the given equation by 5 yields $|x| = 9$. By the definition of absolute value, if $|x| = 9$, then $x = 9$ or $x = -9$. Therefore, the two solutions to the equation $5|x| = 45$ are 9 and -9 . It follows that the positive solution to the given equation is 9.